

Flip-Flops – One Page Study Notes (SR, JK, D, T)

1. SR Flip-Flop (Set–Reset)

Q is set or reset based on S and R inputs.

Truth Table:

S R | Q(next)

0 0 | Q (hold)

0 1 | 0 (reset)

1 0 | 1 (set)

1 1 | Invalid

2. JK Flip-Flop

Improved SR; no invalid state.

Truth Table:

J K Q(current) | Q(next)

0 0 Q | Q (hold)

0 1 Q | 0 (reset)

1 0 Q | 1 (set)

1 1 0 | 1 (toggle)

1 1 1 | 0 (toggle)

3. D Flip-Flop (Data Flip-Flop)

On clock pulse: $Q(\text{next}) = D$

Truth Table:

D | Q(next)

0 | 0

1 | 1

4. T Flip-Flop (Toggle Flip-Flop)

T = 1 toggles output on clock.

Truth Table:

T Q(current) | Q(next)

0 0 | 0 (hold)

0 1 | 1 (hold)

1 0 | 1 (toggle)

1 1 | 0 (toggle)

Summary:

SR – Basic Set/Reset

JK – Set/Reset/Toggle

D – Data stored on clock

T – Toggle on clock (used in counters)